

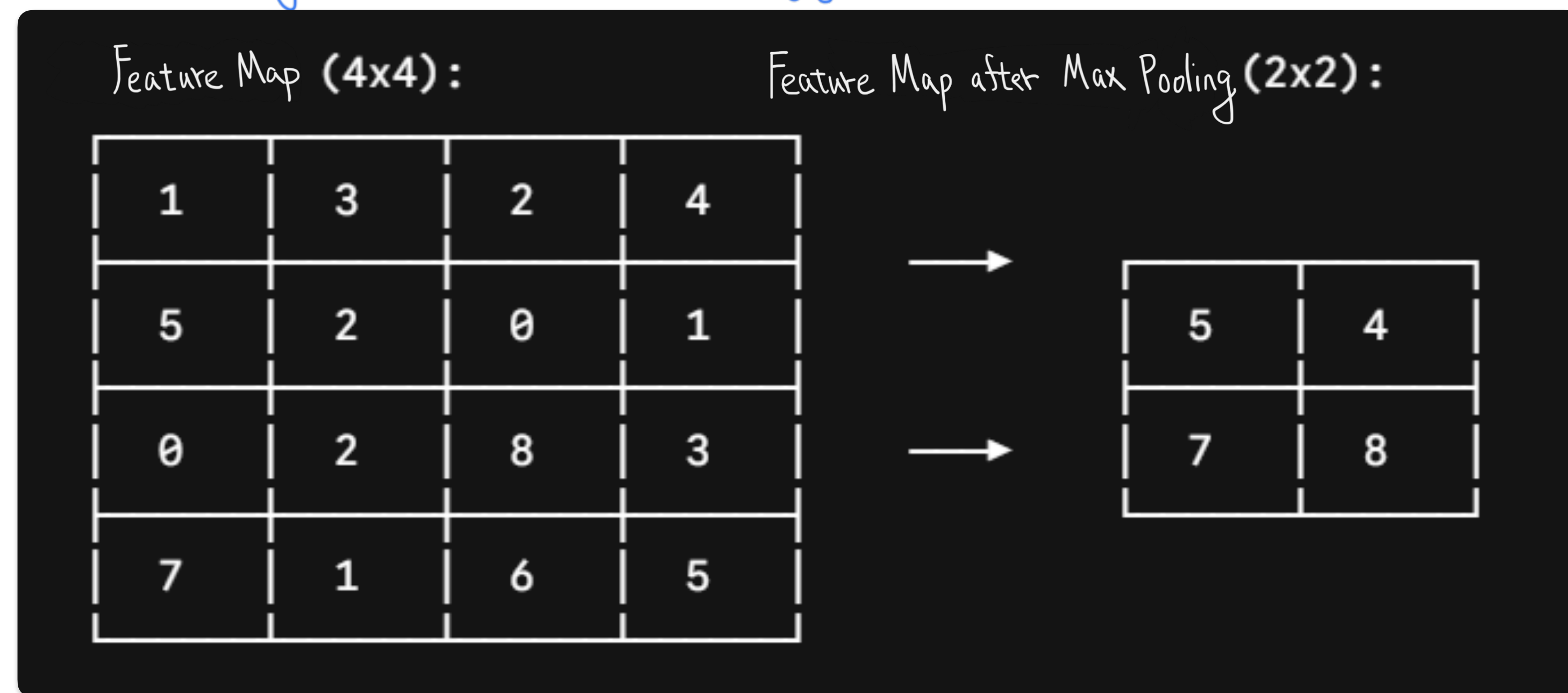
5. Max Pooling:

A. max pooling is a downsampling operation that slides an $N \times N$ pooling window with a specific stride across a feature map, keeping only the maximum value in the region covered by the $N \times N$ pooling window, while discarding the rest of the numbers.

B. a 2×2 pooling window and stride=2:

⇒ from each cluster of 4 neurons, it only extracts the single highest value.

⇒ following the standard topology, this operation cuts both the width and the height of the feature map in half while preserving the total num of feature maps in the convolutional layer.



6. Num of Data Inputs for A Single Neuron in A Convolutional Layer:

$$DI_c = K \times K \times C_{in}, \left[\begin{array}{l} DI_c = \text{num of data inputs for a single neuron in a convolutional layer.} \\ K \times K \times C_{in} = \text{size of a convolutional kernel.} \\ K = \text{the side length of a convolutional kernel as seen in a 2D plane.} \\ C_{in} = \text{num of input channels in a convolutional layer.} \end{array} \right]$$

7. The Fully Connected Layer in a CNN:

A. a CNN doesn't consist solely of convolutional layers,

it is common to append one or more fully connected layers at the end of the neural network to combine all complex features extracted by the final convolutional layer.

B. before passing the features to the fully connected layers, we must first flatten the features extracted by the 3D final convolutional layer into a 1D array ($\equiv$ a vector), because the fully connected layers has/have no concept of spatial dimensions.